

AR RAQQA, Syrian Arab Republic

WMO: 400390

Lat: 35.933N

Lon: 39.017E

Elev: 246

StdP: 98.40

Time Zone: 2.00 (SYR)

Period: 82-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-0.7	-9.1	1.8	3.1	-7.2	2.1	4.3	8.3	10.6	6.2	10.5	0.1	290	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	41.0	22.5	39.3	21.8	37.8	21.3	24.4	37.2	23.5	35.9	22.8	34.7	1.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.6	31.3	19.5	14.7	30.8	18.6	13.8	30.6	75.2	37.4	71.7	36.0	68.8	34.5	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	4.9	4.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-5.0	26.0	1.7	2.1	-6.2	27.5	-7.2	28.8	-8.1	29.9	-9.3	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	7.3	9.6	13.3	18.5	23.4	28.7	31.0	30.6	26.5	21.1	13.6	8.5	(6)
(7)		DBStd	8.93	3.03	3.14	3.65	3.52	2.94	2.35	2.21	2.28	2.60	2.96	3.56	3.02	(7)
(8)		HDD10.0	222	92	42	12	1	0	0	0	0	0	0	11	65	(8)
(9)		HDD18.3	1247	342	246	159	38	2	0	0	0	0	11	145	304	(9)
(10)		CDD10.0	3653	9	29	115	256	416	561	651	640	495	345	118	19	(10)
(11)		CDD18.3	1636	0	0	4	44	159	311	392	382	245	97	2	0	(11)
(12)		CDH23.3	21197	0	0	59	483	1744	4204	5739	5354	2744	846	24	0	(12)
(13)		CDH26.7	11902	0	0	9	167	782	2474	3570	3243	1378	277	1	0	(13)
(14)	Wind	WSAvg	1.2	0.7	1.1	1.2	1.2	1.3	1.8	2.2	1.8	1.1	0.7	0.5	0.5	(14)
(15)	Precipitation	PrecAvg	179	38	29	29	21	12	0	0	1	8	21	25	(15)	
(16)		PrecMax	312	124	82	117	64	101	6	2	0	7	79	102	(16)	
(17)		PrecMin	55	6	8	2	0	0	0	0	0	0	0	4	(17)	
(18)		PrecStd	70	26	19	28	19	21	1	0	0	2	10	20	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	20.2	28.1	34.6	36.5	42.2	43.1	37.5	33.0	25.9	19.4	(19)	
(20)			MCWB	11.8	11.8	16.0	18.2	19.3	21.8	22.9	23.9	22.7	19.0	14.9	10.9	(20)
(21)		2%	DB	15.2	18.4	24.8	30.8	34.5	39.4	41.1	40.7	35.8	31.2	23.8	16.5	(21)
(22)			MCWB	10.3	11.0	13.9	17.2	18.5	20.7	22.1	23.5	21.9	18.8	14.4	10.6	(22)
(23)		5%	DB	13.7	16.9	22.8	28.3	32.7	37.7	39.5	39.0	34.6	29.7	22.2	15.0	(23)
(24)			MCWB	9.5	10.5	13.3	16.3	17.9	19.7	21.5	22.9	21.3	18.5	14.0	10.0	(24)
(25)		10%	DB	12.2	15.1	20.7	26.0	31.0	36.2	37.9	37.4	33.3	28.0	20.4	13.5	(25)
(26)			MCWB	8.6	9.6	12.4	15.2	17.5	19.4	21.2	22.3	20.7	17.8	13.2	9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.3	16.9	19.7	20.7	24.0	25.5	25.8	24.3	21.1	16.6	13.3	(27)
(28)			MCDWB	16.8	18.6	25.0	31.0	33.2	39.2	37.9	38.5	34.8	30.2	22.9	16.2	(28)
(29)		2%	WB	11.3	11.8	15.4	18.1	19.6	22.1	23.9	24.7	23.0	19.9	15.5	11.9	(29)
(30)			MCDWB	14.1	16.9	22.8	28.4	32.1	35.7	37.1	37.6	33.8	29.3	21.8	15.0	(30)
(31)		5%	WB	10.1	11.0	14.1	17.0	18.7	21.2	23.0	23.9	22.3	19.0	14.7	11.0	(31)
(32)			MCDWB	13.0	15.4	20.7	26.9	30.5	34.5	36.0	36.6	32.6	28.1	20.8	13.7	(32)
(33)		10%	WB	9.1	10.3	12.9	15.9	17.9	20.4	22.3	23.2	21.6	18.2	13.7	10.0	(33)
(34)			MCDWB	11.5	14.3	19.4	24.5	29.0	33.4	35.2	35.2	31.5	26.8	19.3	12.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	12.0	12.6	13.6	14.2	13.8	13.6	13.3	13.5	12.4	10.6	(35)
(36)			MCDBR	10.7	12.3	15.2	16.3	15.8	16.3	15.4	15.2	14.5	15.8	14.8	12.0	(36)
(37)			MCWBR	6.1	6.2	7.1	6.8	5.9	5.7	5.5	6.4	6.6	7.2	7.4	6.9	(37)
(38)		5% WB	MCDBR	8.9	10.6	13.2	14.7	14.9	15.0	14.1	14.4	13.7	14.3	12.6	9.9	(38)
(39)			MCWBR	5.5	5.8	6.6	6.6	6.1	5.6	5.5	6.5	6.8	6.9	6.5	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.378	0.407	0.433	0.501	0.494	0.439	0.432	0.439	0.416	0.454	0.392	0.378	(40)	
(41)		taud	2.156	2.070	2.013	1.855	1.906	2.074	2.110	2.085	2.136	2.004	2.137	2.155	(41)	
(42)		Ebn at Noon	789	812	828	791	803	847	849	836	834	750	764	757	(42)	
(43)		Edh at Noon	115	141	163	202	194	164	157	158	143	149	117	109	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.50	3.38	4.80	5.95	7.09	8.12	7.87	7.01	5.84	4.21	3.04	2.28	(44)	
(45)		RadStd	0.20	0.32	0.26	0.34	0.39	0.26	0.16	0.21	0.22	0.23	0.25	0.23	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	-1.77	N/A	-0.89	N/A	N/A	-54	N/A	N/A

Nomenclature: See separate page